

DS Unit 3 Previous Year Questions

Data Structures (DS) · Unit 3 · Trees & Binary Search Trees · 2023

Questions collected from previous year university examinations (2019–2023).

Part A — Short answer (2 marks)

1. Discuss the role of Binary Tree Traversals in Trees & Binary Search Trees. [2023]
2. Define Binary Search Trees. Derive or justify the key properties involved. [2019]
3. Explain AVL Trees with a neat diagram and a suitable example. [2020]
4. Compare Heaps with Threaded Binary Trees. When would you prefer one over the other? [2020]

Part B — Long answer (10 marks)

1. What is meant by B-Trees? Discuss its advantages and limitations. [2019]
2. Write short notes on Threaded Binary Trees. [2021]
3. Write short notes on Binary Tree Traversals. [2021]
4. Write short notes on Binary Search Trees. [2020]
5. Illustrate AVL Trees with a worked example. [2019]
6. Discuss the role of Heaps in Trees & Binary Search Trees. [2020]
7. Discuss the role of B-Trees in Trees & Binary Search Trees. [2022]
8. What is meant by Threaded Binary Trees? Discuss its advantages and limitations. [2020]
9. Compare Binary Tree Traversals with Heaps. When would you prefer one over the other? [2021]
10. Discuss the role of Binary Search Trees in Trees & Binary Search Trees. [2021]